Shortly - URL Shortener

A practical web application for creating and sharing short links

1 NAME AND ROLE APPLIED FOR

Alajingi Shivani

Role applied for: Software Development Engineer (SDE)

2 PROJECT TITLE AND SHORT DESCRIPTION

Shortly - URL Shortener

Shortly is a simple URL shortener for users who want to turn long web addresses into clean, shareable links. It supports generated short codes and optional custom aliases.

3 PROBLEM STATEMENT AND SOLUTION

Long URLs are difficult to share, read, and remember, especially in messages or printed content. Shortly validates HTTP/HTTPS links, creates a unique six-character code or accepts a valid custom alias, and stores the mapping for later use. When a short link is opened, the service looks up the code and redirects the user to the original URL.

4 TECHNOLOGIES USED

Node.js	Runs the local HTTP server and backend request handling.
MongoDB	Stores short-link mappings and rate-limit counters.
JavaScript	Implements API functions and browser form behaviour.
HTML and CSS	Provide the user interface and responsive presentation.
Vercel	Hosts the deployed API functions and application.

5 KEY FEATURES AND MY CONTRIBUTION

- Creates random six-character URL-safe codes or user-selected aliases.
- Validates input, accepts only HTTP/HTTPS URLs, and returns clear API errors.
- Redirects valid short links with HTTP 302 responses.
- Applies a shared limit of 10 create requests per client IP in 60 seconds.
- Provides a browser interface with generated-link display and copy action.

My contribution: I implemented the Node.js backend, API routes, validation, and redirect flow. I designed the MongoDB access helpers, unique-code handling, and database-backed rate limiting, then connected the frontend to the API and prepared Vercel deployment.

6 GITHUB REPOSITORY LINK

https://github.com/shivanialajingi-prog/url-shortner

7 LIVE DEMO LINK

https://url-shortner-tan-iota.vercel.app/